ARIJIT CHOWDHURY

Kolkata,WB,IND | Email:arj.chowdhury.official.03@gmail.com | Phone No: 9007019616 | GitHub: ARI-900 | LinkedIn: Arijit | Blog: Link

SKILLS

- **Languages:** C, C++, Python, JavaScript, TypeScript, Java
- **Frameworks & Libraries:** React, Next.js, Node.js, Express.js, Django, Redux.
- **Databases:** MySQL, MongoDB.
- **DevOps & Tools:** Git, GitHub, Docker, Kubernetes, Microservices, CI/CD, Agile Methodologies.
- **CS Core:** OOPS,SOLID,DBMS,Operating Systems,Computer Networks,DSA, REST API Design,Multithreading,Concurrency.

EXPERIENCE

1. Software Engineering Intern — Space Applications Centre (SAC) **Sep 2025 – Jan 2026**
Indian Space Research Organisation (ISRO) | Ahmedabad, India | [*Link*]

- Designed and trained an **Attention Residual U-Net** to correct systematic biases in numerical ocean model outputs using satellite altimeter observations as ground truth — achieving **122% R2 gain** (Arabian Sea) and reducing global **RMSE from 0.082m to 0.066m** on unseen test data.
- Formulated an end-to-end ML data pipeline benchmarking model predictions against 3 numerical ocean models across 2 regional zones, incorporating **bilinear interpolation, grid alignment, and anomaly isolation** to evaluate improvement over baseline forecasts across the North Indian Ocean.
- Engineered **hybrid loss functions** and **AI-assisted Python pipelines**, using **SSIM/PSNR metrics** to optimize efficiency by 25%.

PROJECTS

1. NovaHTTP — Multithreaded C++ HTTP Server **Jun 2026 - Aug 2026**
Technologies: C++17, POSIX Sockets, Multithreading, Docker, CMake | *[GitHub]*

- Built a **multithreaded HTTP server** from scratch in C++ using raw **POSIX sockets**, featuring a custom request parser, response builder, and **segment-based router** supporting 6 REST endpoints with path parameters.
- Implemented a variable-size **thread pool** utilizing **mutex**, **condition variable** synchronization for safe concurrent connection handling, benchmarked system limits via **Apache Bench**, sustaining **280+ RPS** across **1,000 requests to 50 concurrent connections**, zero failures and **sub-180ms mean response time**.
- Configured thread-safe singleton logging with per-request execution-time tracking and JSON-driven runtime configuration, and containerized the build-to-run pipeline with Docker and CMake for one-command reproducible deployment.

2. DSA Visualizer — Algorithm Visualization Platform **Feb 2025 – Mar 2025**
Technologies: React, JavaScript, Tailwind CSS, React Router | *[GitHub] [Live]*

- Engineered a non-blocking, step-driven visual execution engine leveraging **JavaScript Generator Functions** to handle complex asynchronous state tracking and real-time visualization across **7 core algorithms**.
- Optimized **React DOM reconciliation paths** with tailored custom hooks, mitigating redundant component re-renders by **40%** to deliver a fluid, theme-aware **60 FPS** user interface.

3. Tech Blogging App — Backend **Oct 2024 – Dec 2024**
Technologies: Node.js, Express.js, MongoDB, JWT | *[GitHub]*

- Architected an MVC-compliant **RESTful API** handling **15+ functional endpoints** with multi-layered security, utilizing **httpOnly cookies**, **JWTs**, and **bcrypt** password hashing algorithms.
- Formulated a **2-layer security middleware chain** for role-based access control (**RBAC**) and optimized indexing strategies on **MongoDB** queries to accelerate multi-field search and category filtering speeds.

4. C++ DSA Library — AlgoSmith **Feb 2026 – Mar 2026**
Technologies: C++17, Templates, CMake, GoogleTest | *[GitHub] [Live]*

- Engineered a header-only **C++17 template library** implementing custom sequential and hierarchical data structures with strict **O(1) tail bounds** and **O(log n) search complexities**.
- Architected a decoupled structure separating **.hpp and .tpp files**, enforcing **Rule of Five** and **RAII** for leak-free memory, validated via **automated unit testing** with GoogleTest across **93/93 passed test cases**.

ACHIEVEMENTS

- Solved 600+ DSA problems across LeetCode and GeeksforGeeks (GFG), covering Stack, Queue, Graphs, Trees, and Dynamic Programming.
- Built an IoT-based smart fire detection system delivering real-time monitoring and automated emergency response in under 2 seconds. published in IoT Insights: Design and Build IoT Applications (Ch. 2) — [View Chapter PDF].

EDUCATION

- B.Tech in CSE-AIML with CGPA: 8.2, MAKAUT University – 2026